



Specialization

- Top-down design process; we designate sub-groupings within an entity set that are distinctive from other entities in the set.
- These sub-groupings become lower-level entity sets that have attributes or participate in relationships that do not apply to the higher-level entity set.
- Depicted by a *triangle component labeled ISA* (e.g., *instructor “is a” person*).
- **Attribute inheritance** – a lower-level entity set **inherits all the attributes** and relationship participation of the higher-level entity set to which it is linked.
- The process of designating subgroupings within an entity set is called **specialization**.



Module 2

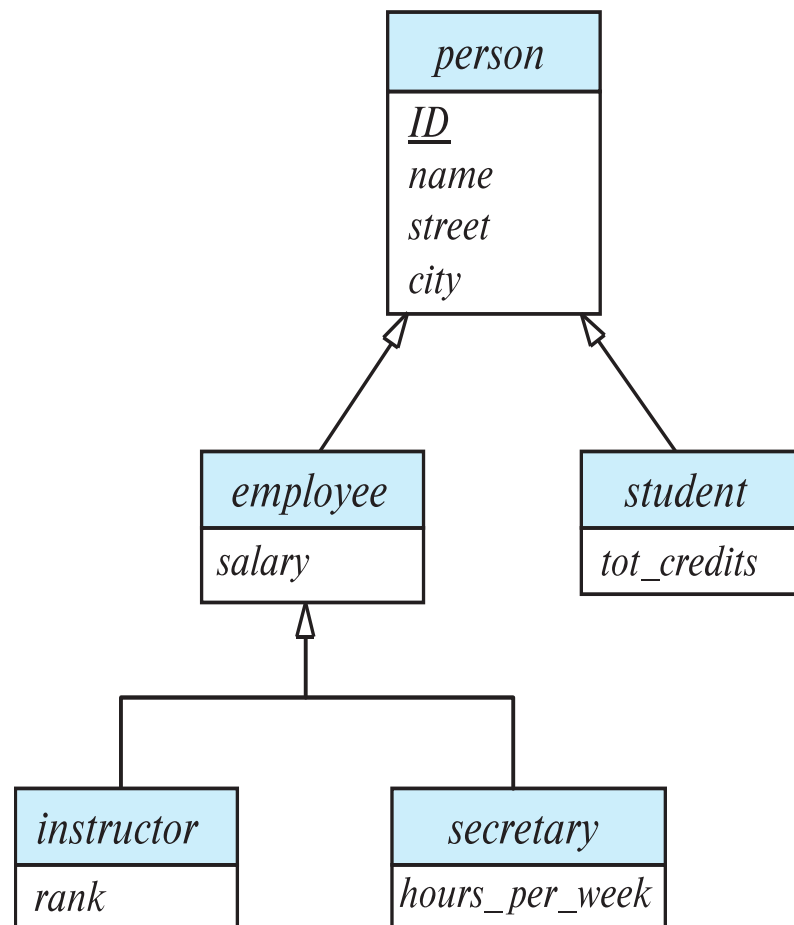
Relational Model and E-R Modeling

Topic:
Extended E-R Features
(Silberschatz)



Specialization Example

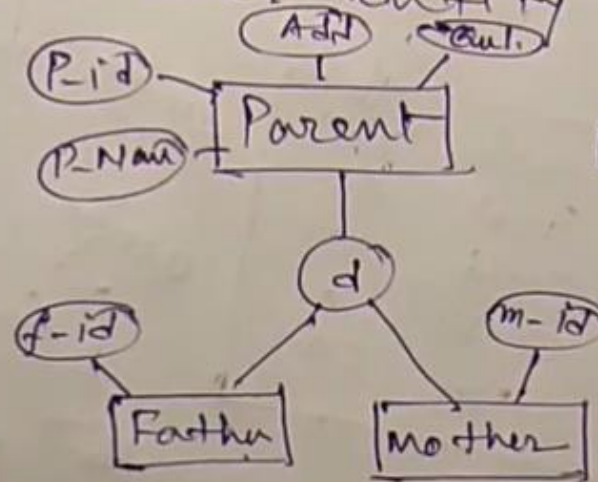
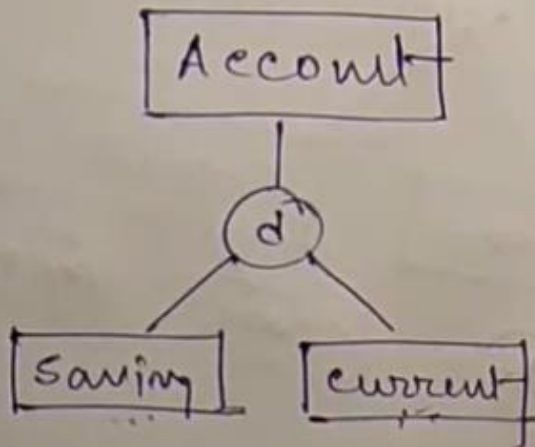
- **Overlapping** – *employee* and *student*
- **Disjoint** – *instructor* and *secretary*
- The specialization relationship may also be referred to as a **superclass-subclass relationship**.





Specialization Example

(a) Disjoint: An entity belong to no more than one lower level entity set;



Parent(P-id, P-Name, Add, Email.)

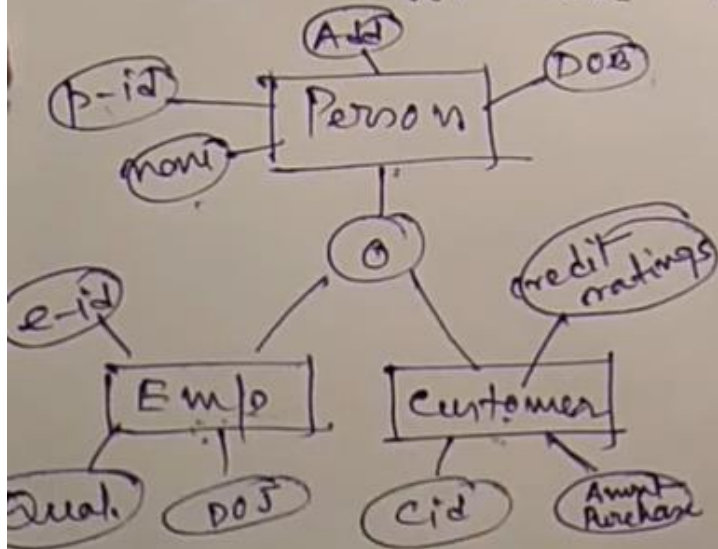
Father(f-id, —, —)

Mother(m-id, —, —)



Specialization Example

(b) Overlapping: The same entity may belong to more than one lower-level entity set within a single generalization.



Person (P-id, Name, Add, DOB)
EMP (e-id, Qualification, DOJ)
customer (cid, Amount Purchase, credit ratings)



Representing Specialization via Schemas

■ Method 1:

- Form a schema for the higher-level entity
- Form a schema for each lower-level entity set, include primary key of higher-level entity set and local attributes

schema	attributes
person	ID, name, street, city
student	ID, tot_cred
employee	ID, salary

- **Drawback:** getting information about, an *employee* requires accessing two relations, the one corresponding to the low-level schema and the one corresponding to the high-level schema



Representing Specialization as Schemas (Cont.)

■ Method 2:

- Form a schema for each entity set with all local and inherited attributes

schema	attributes
person	ID, name, street, city
student	ID, name, street, city, tot_cred
employee	ID, name, street, city, salary

- Drawback: *name*, *street* and *city* may be stored redundantly for people who are both students and employees

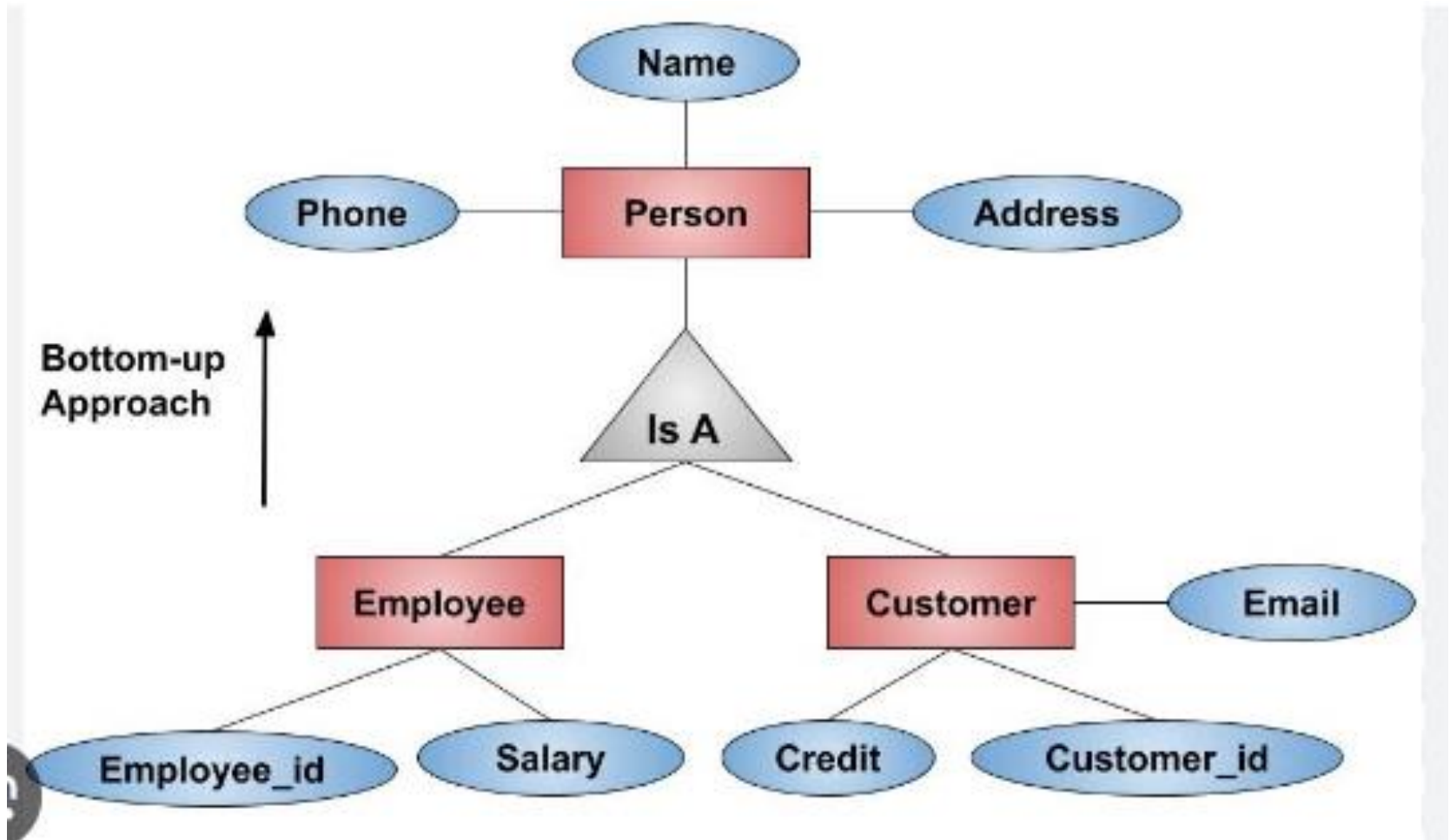


Generalization

- **A bottom-up design process** – combine a number of entity sets that share the same features into a higher-level entity set.
- Specialization and generalization are **simple inversions of each other**; they are represented in an E-R diagram in the same way.
- The terms specialization and generalization are **used interchangeably**.



Generalization





Attribute Inheritance

- A crucial property of the higher and lower-level entities created by specialization and generalization is **attribute inheritance**.
- The attributes of the higher-level entity sets are said to be **inherited by the lower-level entity sets**. For example, *student and employee inherit the attributes of person*.
- A lower-level entity set (or subclass) also **inherits participation in the relationship sets** in which its higher-level entity (or superclass) participates.
- Like attribute inheritance, **participation inheritance applies through all tiers of lower-level entity sets**.



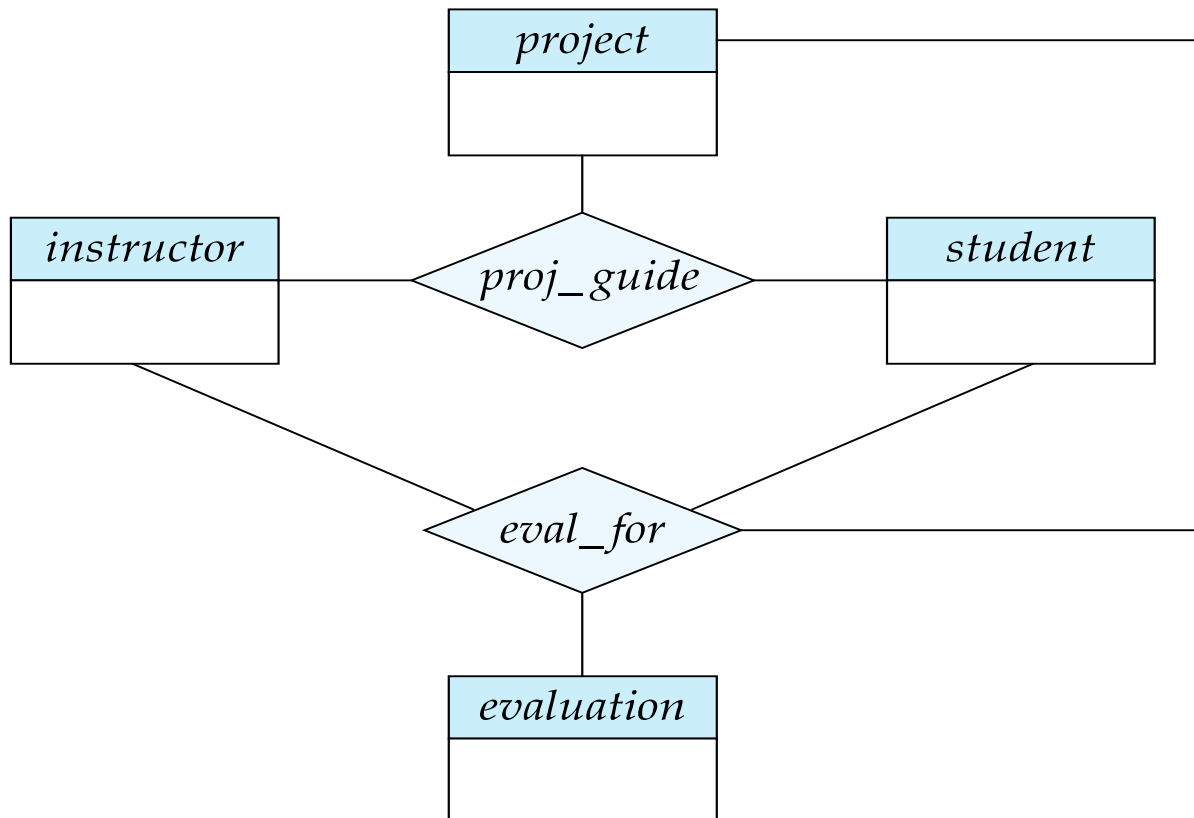
Attribute Inheritance

- A higher-level entity set with attributes and relationships that **apply to all of its lower-level entity sets.**
- Lower-level entity sets with distinctive features that **apply only within a particular lower-level entity set.**



Aggregation

- Consider the **ternary relationship** *proj_guide*, which we saw earlier
- Suppose we want to record evaluations of a student by a guide on a project





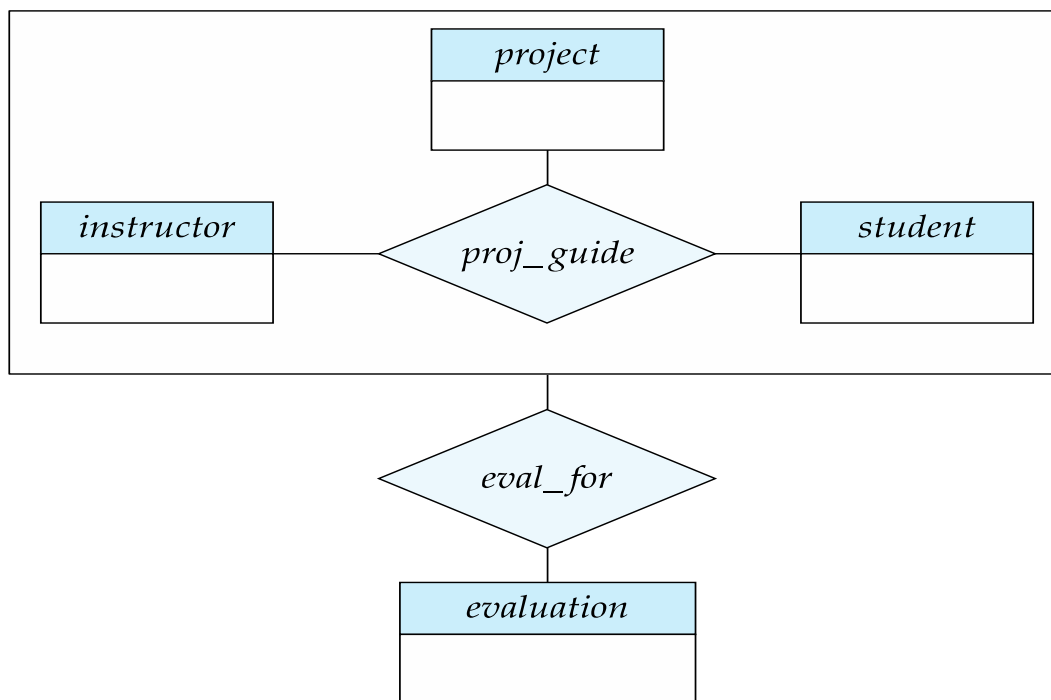
Aggregation (Cont.)

- Eliminate this redundancy via *aggregation*
- **Aggregation** is an abstraction through which relationships are treated as higher-level entities.
 - Treat relationship as an abstract entity
 - Allows relationships between relationships
 - Abstraction of relationship into new entity



Aggregation (Cont.)

- Eliminate this redundancy via *aggregation* without introducing redundancy, the following diagram represents:
 - A student is guided by a particular instructor on a particular project
 - A student, instructor, project combination may have an associated evaluation



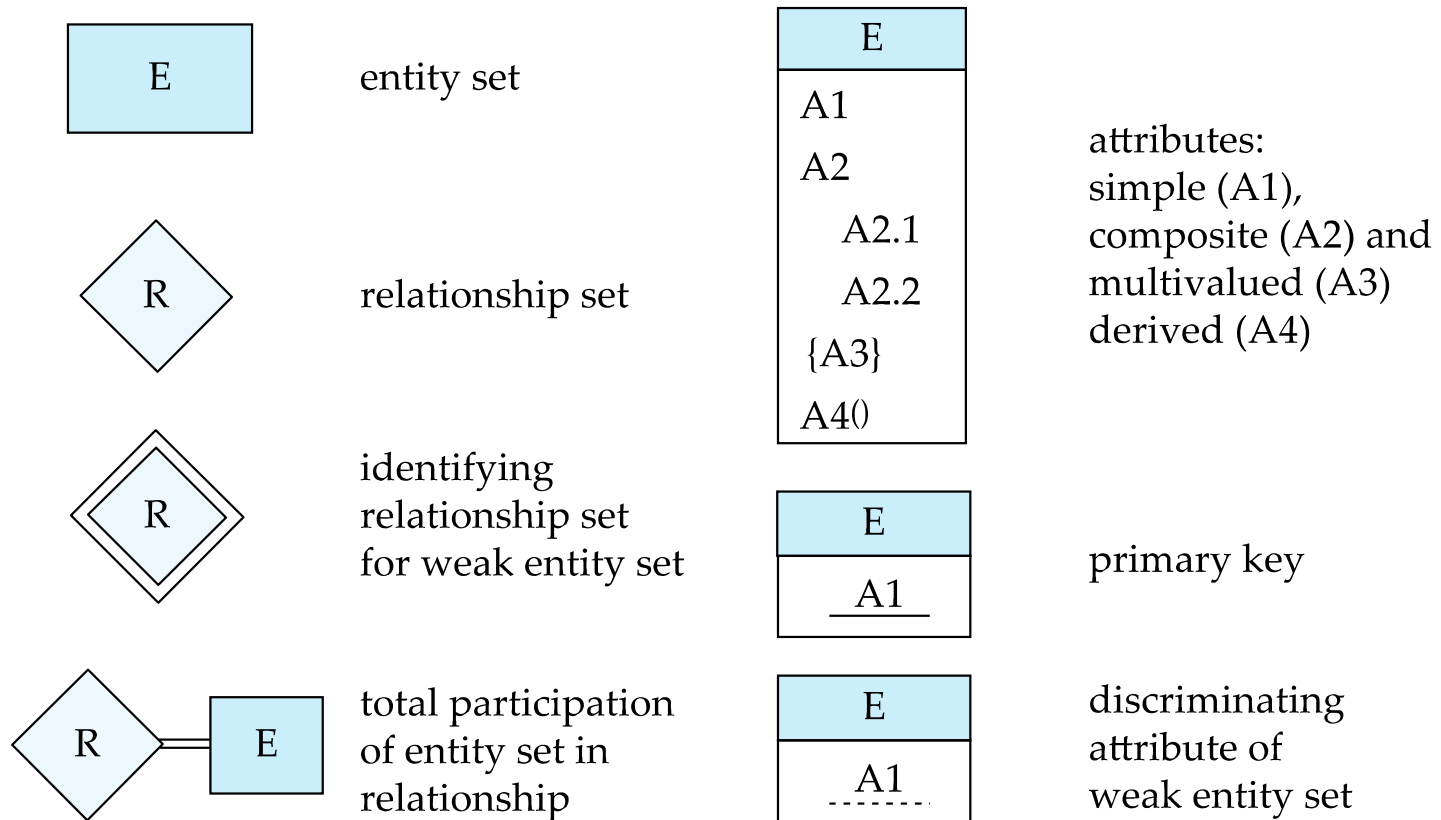


Reduction to Relational Schemas

- To represent aggregation, create a schema containing
 - Primary key of the aggregated relationship,
 - The primary key of the associated entity set
 - Any descriptive attributes
- In our example:
 - The schema *eval_for* is:
eval_for (s_ID, project_id, i_ID, evaluation_id)
 - The schema *proj_guide* is redundant.

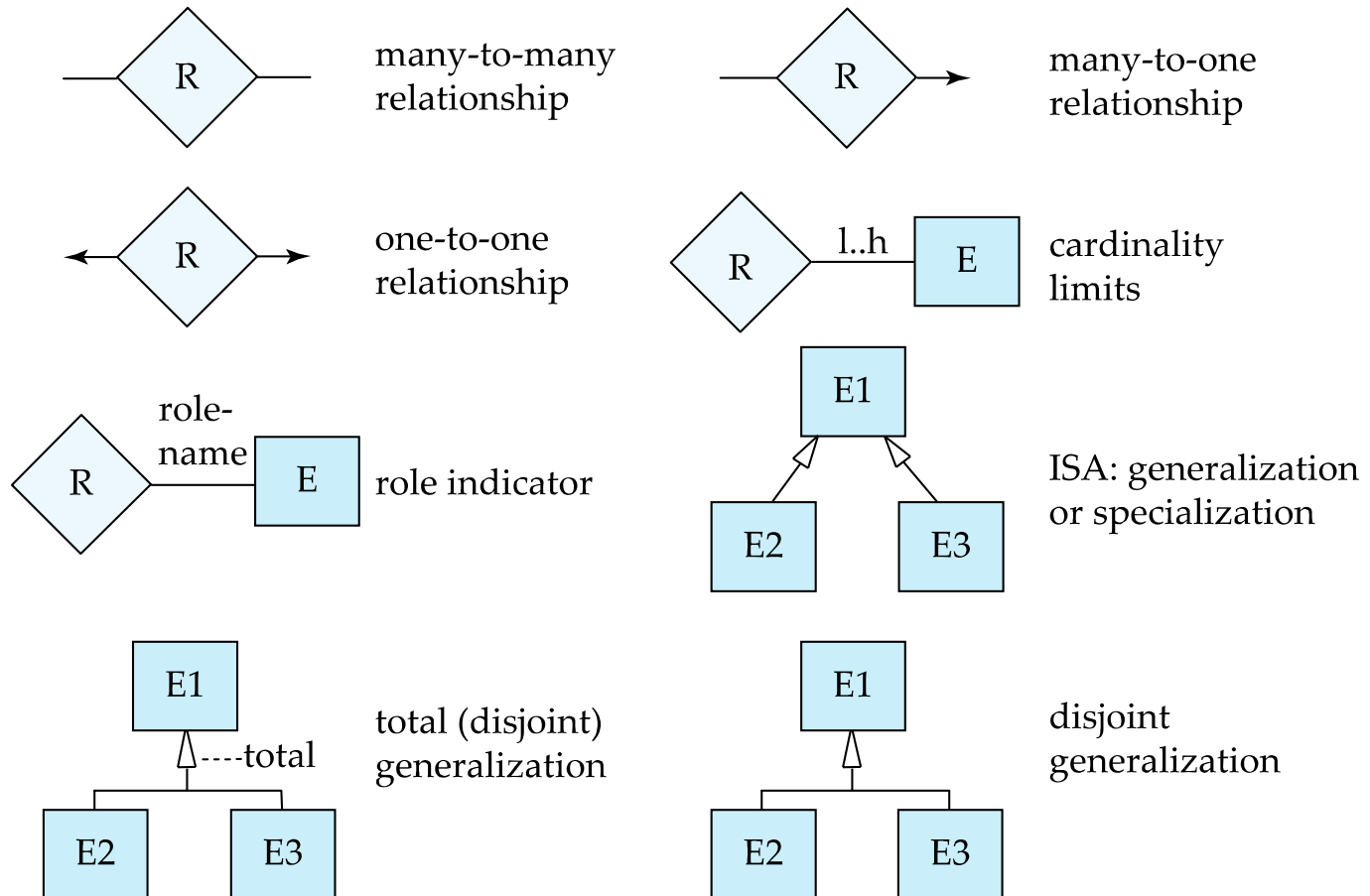


Summary of Symbols Used in E-R Notation





Symbols Used in E-R Notation (Cont.)

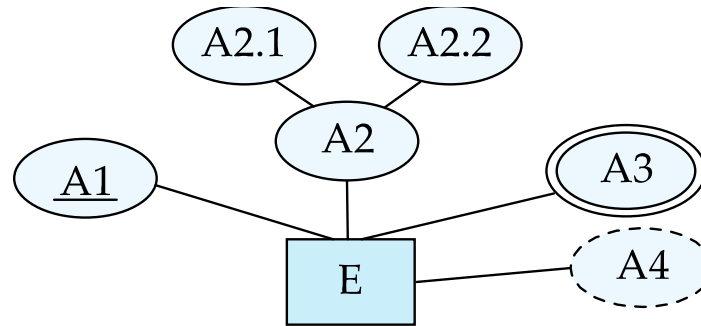




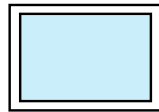
Alternative ER Notations

- Chen, IDE1FX, ...

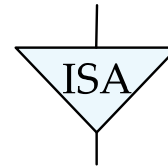
entity set E with
simple attribute A1,
composite attribute A2,
multivalued attribute A3,
derived attribute A4,
and primary key A1



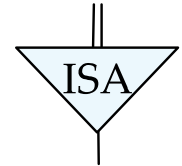
weak entity set



generalization



total
generalization



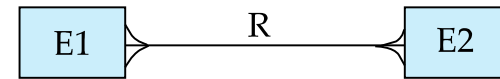
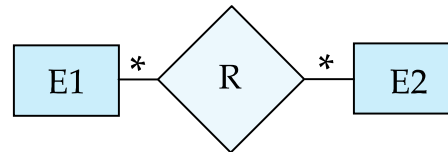


Alternative ER Notations

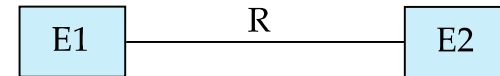
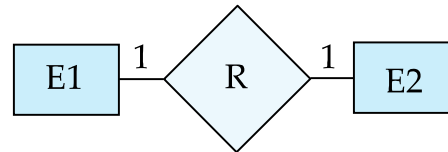
Chen

IDE1FX (Crows foot notation)

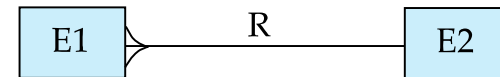
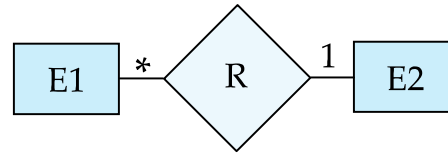
many-to-many
relationship



one-to-one
relationship



many-to-one
relationship



participation
in R: total (E1)
and partial (E2)

